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END SEMESTER EXAMINATIONS FALL 2022

Faculty	FoCAS	Department	Computer science	Batch	2020(F)
Semester	6^{th}	Degree Program	BSCS	Course Instructor	Mr.Noman Uddin Khan/Ms. Nida Jamil, Mr .Muhammad Ali
Course Code	MS203	Course Title	Numerical Analysis	Credit Hours	3
Date of Examinations	04-02-2023	Timings	2:00 – 5:00	Shift	E

Timings: 3 Hours

Max Marks: 50

Instructions:
- Please don't write anything on the Question Paper except for your Roll No. and Name.
- The paper must be solved / attempted on Printed Answer books provided by the Examinations Department.
- Please return the Question Paper along with the Answer books.

Roll No.	215	Name	A. Pehn	Section	E	'A' Copy Serial No.	135263

Q.No.1 (07 Marks)

Using Taylor's series method, find the solution of the initial value problem within [1,1.2] of order 4 $\frac{dy}{dx} = x + y$ $y(1) = 0$ at $x = 1.2$, with $h = 0.2$

Q.No.2 (07 Marks)

Use R-K fourth order methods to numerically solve the following initial value problem

$$\frac{dy}{dx} = 100\ e^{-100(x-1)^2}, \quad y(0.8) = 4.9491 \text{ and find } y(0.9)$$

Q.No.3 (07 Marks)

Find solution of $y' = \frac{x+y}{2}$ at y (2) by Milne's Predictor corrector formula when y (0) =2, y (0.5) =2.636, y (1) = 3. 595 and y (1.5) =4.968

Q.No.4 (07 Marks)

Find solution at x = 6 by using Newton's Divided difference interpolation formula

X	f(x)
5	150
7	392
11	1452
13	2366
21	9702

$\frac{1}{6}\{k+k+k+k\}$

192 $\frac{8}{7336}$ $\frac{14}{\frac{460}{10}}$

$\frac{242}{2}$ $\frac{1060}{4}$ 914

2.44780

$y_0 + \frac{4h}{3}(2f_0 - f_1 + f_2)$

$y_2 + \frac{h}{3}(f_2 + 4f_3 + f_4)$.

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Q.No.5 **(07 Marks)**

Using Lagrange's interpolation formula find y (55) from the following table:

X	50 x_0	52 x_1	54 x_2	56 x_3	58 x_4
Y	3.684	3.732	3.779	3.825	3.870

Q.No.6 **(10 Marks)**

Solve the integral $\int_0^1 x^2\,dx$ into four subintervals by trapezoidal rule and Simpson 1/3 Rule. Compare your answer with its true value and state which rule is more appropriate.

Q.No.7 **(05 Marks)**

Use Bisection method to find its positive root of the equation $xe^x = 1$ (up to five iterations)

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Faculty	FoCAS	Department	CS&IT	Batch	2020S
Semester	6th	Degree Program	BSCS	Course Instructor	Ayesha Khaliq
Course Code	CS425	Course Title	Design and Analysis of Algorithm	Credit Hours	3+0
Date of Examinations	7/2/2023	Timings	2:00-5:00	Shift	Evening

Timings: 3 Hours

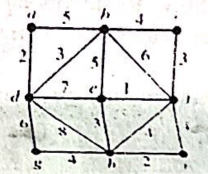
Max Marks: 50

Instructions:
- Please don't write anything on the Question Paper except for your Roll No. and Name.
- The paper must be solved / attempted on Printed Answer books provided by the Examinations Department.
- Please return the Question Paper along with the Answer books.

Roll No.	215	Name	ABDUL Roh	Section E		'A' Copy Serial No.	141751

Q.No.1 (7+3)marks

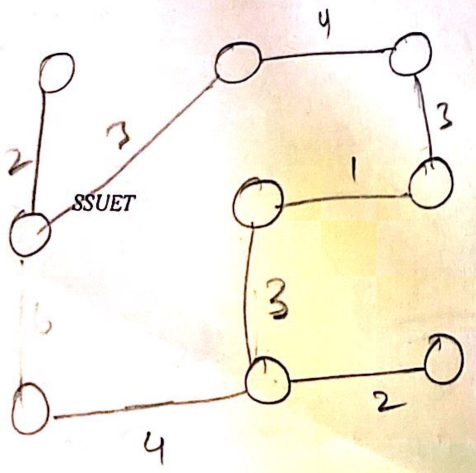
(a) Use Prim's algorithm to find minimum spanning tree for the given weighted graph.

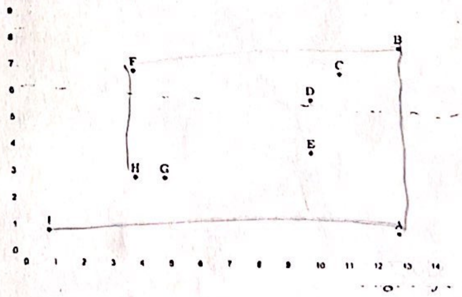


(b) Write the differences of greedy and divide conquer approach with the help of example.

Q.No.2 10 marks

Run the Graham scan algorithm to compute the convex hull of points below, starting from A. Give the points that appear on the trial hull (after each iteration) in the order that they appear.





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Q.No.3 **(5+5) marks**

a) Describe the method of reading excel and csv files in python and explain what operation can be performed on it.

b) From the given dataset do step by step exploratory data analysis report. Do at least any five operation on the dataset.

	ID	Name	Sex	Age	Height	Weight	Team	NOC	Games	Year	Season	City	Sport	Event	Medal
0	1	A Dijiang	M	24.0	180.0	80.0	China	CHN	1992 Summer	1992	Summer	Barcelona	Basketball	Basketball Men's Basketball	NaN
1	2	A Lamusi	M	23.0	170.0	60.0	China	CHN	2012 Summer	2012	Summer	London	Judo	Judo Men's Extra Lightweight	NaN
2	3	Gunnar Nielsen Aaby	M	24.0	NaN	NaN	Denmark	DEN	1920 Summer	1920	Summer	Antwerpen	Football	Football Men's Football	NaN
3	4	Edgar Lindenau Aabye	M	34.0	NaN	NaN	Denmark/Sweden	DEN	1900 Summer	1900	Summer	Paris	Tug-Of-War	Tug-Of-War Men's Tug-Of-War	Gold
4	5	Christine Jacoba Aaftink	F	21.0	185.0	82.0	Netherlands	NED	1988 Winter	1988	Winter	Calgary	Speed Skating	Speed Skating Women's 500 metres	NaN

Q.No.4 **(8+2) marks**

a) Find the pattern of given text using Rabin Karp algorithm. Show each step to find correct string. **String="2022SPRING" and Pattern="RING"**

b) Justify the reason for worst case time complexity in Rabin Karp algorithm which is same of brute force search.

Q.No.5 **10 marks**

Solve the following Knapsack 1/0 problem by dynamic programming. Show calculation for each iteration on the table. The total Capacity is W=5

[illegible]	1	2	3	4
w	3	2	4	1
v	8	3	9	6

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lty	FoCAS	Department	CS&IT	Batch	2020S
nester	6th	Degree Program	BS(CS)	Course Instructor	Ms. Rabiya Tahir & Ms. Quratulain Khalid
Course Code	CS-325T	Course Title	Artificial Intelligence	Credit Hours	3
Date of Examinations	31-01-2023	Timings	02:00pm to 05:00pm	Shift	2nd

Timings: 3 Hours

Max Marks: 50

Instructions:

- Please don't write anything on the Question Paper except for your Roll No. and Name.
- The paper must be solved / attempted on Printed Answer books provided by the Examinations Department.
- Please return the Question Paper along with the Answer books.

Roll No.	215	Name	ABDUL REHMAN	Section B		'A' Copy Serial No.	124712

Q.No.1 (4+2+2+2) Marks

(a) Convert the following into first order logic.

i. Every student walks or talks.
ii. If anyone cheats, everyone suffers.
iii. Lucy criticized John.
iv. Not all cars have carburetors.

(b) Show that the two statements $(p \wedge q) \rightarrow r$ and $(p \rightarrow r) \wedge (q \rightarrow r)$ are not logically equivalent.

(c) Write down the given statement in propositional logic its converse, contrapositive and inverse.

i. If I had $1,000,000, I'd buy you a fur coat. 4 if P→Q ¬P→¬Q

(d) Show that $[(p \vee q) \wedge (r \vee \neg q)] \rightarrow (p \vee r)$ is a tautology by making a truth table.

Q.No.2 (7+2+1) Marks

Consider an image shown in Fig: 01 perform the following operations to extract the feature vector.

(a) Derive an activation map by convolving an image with the kernel having stride size=~~2~~. 1
(b) Apply max pooling on an activation map which has been derived in part (a). — strid = 2
(c) Apply flattening on the matrix derived in part (b). — Transpose

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

-1	1	-1
-1	1	-1
-1	1	-1

Kernel

Fig: 01

∃ one P
All P
∨ one T
all T

if T→F=R

Q.No.3 (2+2+5+1) Marks

Multilayer perceptron architecture is given in Fig: 02 having inputs **x1=0.2 and x2=0.4** with initial weights **[w13, w23, w14, w24, w35, w45]** i.e. **[0.1, 0.9, 0.5, 0.8, 0.7, 0.5]**given in diagram. The target output for **a5is 1**. The network produces the following outputs in 1st iteration. a4= 0.603, a3= 0.594 and a5= 0.672. Given that learning rate **α=1**, activation function 'g' is **sigmoid**. Calculate the following:

(i) Error factor.
(ii) Adjust the weights.

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(iii) Compute output with new weight values.
(iv) Compare the output with 1st iteration and give your observations.

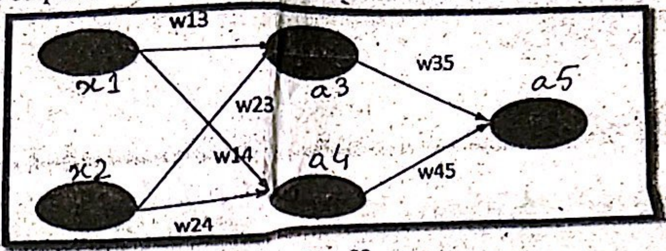

Fig: 02

(8+2) Marks

Q.No.4

(a) In Bayesian spam filtering, an e-mail message is modeled as an unordered collection of words selected from one of two probability distributions: one representing **"spam"** and one representing legitimate e-mail (**"not spam"**). Imagine there are two literal bags full of words. One bag is filled with words found in spam messages, and the other with words found in legitimate e-mail. While any given word is likely to be somewhere in both bags, the "spam" bag will contain spam-related words significantly more frequently, while the "ham" bag will contain more words related to the user's friends or workplace. To classify an e-mail message, the Bayesian spam filter assumes that the message is a pile of words that has been poured out randomly from one of the two bags. Apply naïve bayes theorem on the data shown in Fig: 03 to classify the given email as "spam" or **"not spam"** by calculating the Bayesian probability of **Word=New offer.**

Word	Email
Greetings	Spam
New offer	Not Spam
Greetings	Not Spam
New offer	Spam
Hi	Spam
Greetings	Spam
Hi	Spam
New offer	Not Spam
Greetings	Not Spam
Greetings	Spam
New offer	Not Spam
Hi	Spam
New offer	Spam
Hi	Spam

NO = 5/14
G = 5/14
Hi = 4/14

Fig: 03

(b) Which of the following examples are belong to linear regression and logistic regression?
(i) Whether a person is likely to be infected with COVID-19 or not?
(ii) Prediction of winning in cricket match.

(2+2+2+2+2) Marks

Q.No.5

The two fuzzy sets MF1 and MF2 shown in Fig: 04 having trapezoidal membership function with elements given that MF1 is symmetric and MF2 is asymmetric. MF1= (x1, x3, x5, x7) and MF2= (x4, x6, x8, x9). Where for MF1 (a=0, b=2, c=5, d=7) and for MF2 (a=3, b=7, c=9, d=10). Determine the following:

(i) The membership values of MF1
(ii) The membership values of MF2
(iii) Union of MF1 and MF2 seq
(iv) Intersection of MF1 and MF2 same sap
(v) Complement of MF1 & MF2

same neg.

$$\mu(x) = \max\left(\min\left(\frac{x-a}{b-a}, 1, \frac{d-x}{d-c}\right), 0\right)$$

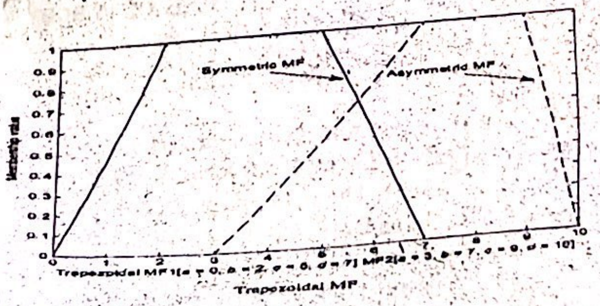

Fig: 04

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Faculty	Computing and Applied Sciences	**Department**	Computer Science & Information Technology	**Batch**	2020S
Semester	6th	**Degree Program**	BS(CS)	**Course Instructor**	Mr.Faraz Khan & Mr.Faheem
Course Code	CS-323	**Course Title**	Computer Graphics and Multimedia Applications	**Credit Hours**	3+1
Date of Examinations	10-2-2023	**Timings**	2.30pm to 5.30pm	**Shift**	Evening

Timings: 3 Hours

Max Marks: 50

Instructions:
- Please don't write anything on the Question Paper except for your Roll No. and Name.
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Roll No.	215	**Name**	A-Rehman	Section		**'A' Copy Serial No.**	

Q.No.1 (5+5) Marks

4 (a) Discuss sound, types , unit , characteristics of sound waves. Also highlight bit depth and bit rate. Elucidate the concept of ADC and DAC with diagram.

(b) How streaming audio speedup data, discuss advantages and disadvantages of streaming with diagram. Discuss web optimization methods for digital audio. Also explain different audio formats in detail.

Q.No.2 (5+5) Marks

(a) Solve and compress the given data by using Huffman encoding Tree Algorithm.

B C C A B B D D A E C C B B A E D D C C

13 6

(b) Discuss pros and cons of GIF, PNG, JPEG file format. Highlight some tips to optimize web graphics.

8 24 48.

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Q.No.3	(4+6) Marks

(a) Write html tag to insert digital audio in background of web page. How much memory is required to store the file size of 30 seconds of 16-bit, 11khz stereo audio.

(b) A square having vertices A(0,0), B(2,0),C(2,2) and D(0,2). Calculate coordinates of square after **translation** when translation factor tk = 3 and translation factor ty = 3. Draw graph before translation and after translation.

Q.No.4	(2+2+6) Marks

(a) Discuss the features of digital audio editing like Trimming, splicing and assembly, volume - adjustment, format conversion, re-sampling and down sampling, equalization, time stretching reversing sound etc.

(b) List down any three free lancing website for graphics designer, also list any three software or tools with respect to image editing, audio editing, video editing, animation, web page designing.

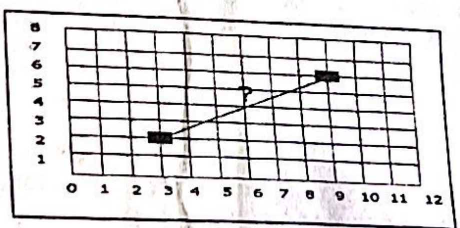
(c) A square having vertices A(0,0), B(2,0), C(2,2) and D(0,2). Calculate coordinates of square **after scaling when scaling factors** sx = 4 and sy = 4. Draw graph before ~~translation~~ Scaling and after **translation.**

Q.No.5	(3+3+4) Marks

(a) Discuss basic principles of animation, Briefly explain the importance of each feature in animation.

(b) Name key points of multimedia project design and development. Also discuss four types of navigation structure with diagram, and give any two examples of multimedia project of every type.

(c) From the below diagram, find out intermediate pixels to turn on using both DDA Algorithm or incremental line algorithm (calculate next 5/6 point/pixels).



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Faculty	Computing and Applied Sciences	**Department**	Computer Science and Information Technology	**Batch**	2020S
Semester	6^{th}	**Degree Program**	BS (CS)	**Course Instructor**	Dr. Amir Mehmood
Course Code	CS-326	**Course Title**	Information Security	**Credit Hours**	03
Date of Examinations	02-02-2023	**Timings**	02:00 PM to 05:00 PM	**Shift**	Evening

Timings: 3 Hours | Max Marks: 50

Instructions:
- Please don't write anything on the Question Paper except for your Roll No. and Name.
- The paper must be solved / attempted on Printed Answer books provided by the Examinations Department.
- Please return the Question Paper along with the Answer books.
- Scientific Calculator are allowed. Show all the steps of calculation on answer script.

Roll No.	215	**Name**	ABDUL REHMAN	Section	E	**'A' Copy Serial No.**	123705.

Q.No.1 (5+5)

a) Suppose you want to enhance 8 bits, or 64 bits block cipher algorithm towards 128 bits cipher. Which changes could be made in the key generation and encryption models.

b) Considering round 2 of the AES Algorithm, Find the word W_8, using AES key expansion where:

W_4	W_5	W_6	W_7
8B	F8	91	E3
FE	96	E5	8C
F1	92	F0	9E
29	43	2C	4B

Q.No.2 (5+5)

D a) Consider two random numbers a = 3 and p = 13, test the usability of these numbers using Fermat's theorem.

b) Perform Encryption and Decryption using the RSA algorithm. p = 5, q = 13, M = 4

Q.No.3 (5+5)

P a) Use the linear congruential method to generate a sequence of five two-digit random integers and corresponding random numbers. Let X_0 = 27, a = 8, c = 47, and m = 100. How we can cover the weaknesses of linear congruential generator.

E[PRauth(T, ID, Pua)

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b) Alice and Bob use the Diffie–Hellman key exchange technique with a common prime q = 157 and a primitive root α = 5.

i. If Alice has a private key $X_A = 15$, find her public key Y_A.

ii. If Bob has a private key $X_B = 27$, find his public key Y_B.

iii. What is the shared secret key between Alice and Bob? Validate by calculating at both ends.

Q.No.4	(5+5)

a) Stronger security for public-key distribution can be achieved by providing tighter control over the distribution of public keys from the directory. Discuss how public keys are distributed among the users? Explain with help of a diagram. Define the role of the master key in this scenario.

$V_1 = \alpha^m \bmod q$

$S_1 = \alpha^K \bmod q$

$S_2 = K^{-1}(m - S_1 X_A) \bmod q-1$

$V_2 = (Y_A)^{X_A} * (S_1)^{S_2} \bmod q$

b) Digital signatures are used for verification of the message initiator by the receiver. Alice wants to sign a message M with the Elgamal digital signature scheme where Alice chooses global parameter q=11, α = 2, her private key = 8, a hash value m = 5 of message M, and K=9. Calculate Alice's public key, the signature pair (S_1, S_2), and how Bob can verify the validation of Alice's signature.

Q.No.5	(5+5)

a) You require confidentiality as well as a digital signature for your website. How will you attain both using a cryptographic hash function? Explain with the help of a diagram. Also discuss the basic differences between Hash and MAC.

b) A client wants a service from a host/server, what procedure /mechanism (s)he has to follow to access the said service.

AES S-Box→

x \ y	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0	63	7C	77	7B	F2	6B	6F	C5	30	01	67	2B	FE	D7	AB	76
1	CA	82	C9	7D	FA	59	47	F0	AD	D4	A2	AF	9C	A4	72	C0
2	B7	FD	93	26	36	3F	F7	CC	34	A5	E5	F1	71	D8	31	15
3	04	C7	23	C3	18	96	05	9A	07	12	80	E2	EB	27	B2	75
4	09	83	2C	1A	1B	6E	5A	A0	52	3B	D6	B3	29	E3	2F	84
5	53	D1	00	ED	20	FC	B1	5B	6A	CB	BE	39	4A	4C	58	CF
6	D0	EF	AA	FB	43	4D	33	85	45	F9	02	7F	50	3C	9F	A8
7	51	A3	40	8F	92	9D	38	F5	BC	B6	DA	21	10	FF	F3	D2
8	CD	0C	13	EC	5F	97	44	17	C4	A7	7E	3D	64	5D	19	73
9	60	81	4F	DC	22	2A	90	88	46	EE	B8	14	DE	5E	0B	DB
A	E0	32	3A	0A	49	06	24	5C	C2	D3	AC	62	91	95	E4	79
B	E7	C8	37	6D	8D	D5	4E	A9	6C	56	F4	EA	65	7A	AE	08
C	BA	78	25	2E	1C	A6	B4	C6	E8	DD	74	1F	4B	BD	8B	8A
D	70	3E	B5	66	48	03	F6	0E	61	35	57	B9	86	C1	1D	9E
E	E1	F8	98	11	69	D9	8E	94	9B	1E	87	E9	CE	55	28	DF
F	8C	A1	89	0D	BF	E6	42	68	41	99	2D	0F	B0	54	BB	16

0B

5

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